ACADEMIC CENTRAL

PRACTICE EXAMINATION

N6

ELECTROTECHNICS N6

PRACTICE EXAM

MARKS: 100 **TIME: 3 hours**

NATED Engineering Studies (Electrical), N6. DC machines, three-phase AC circuits and power factor correction, transformers, three-phase induction motors and synchronous machines. A formula sheet is attached.

Written by Academic Central for practice, following the NATED (Report 191) syllabus for N6. It is not a Department of Higher Education and Training examination paper.

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Show ALL formulae, substitutions and calculations. Write units with all answers.
5. Sketches must be neat, large and fully labelled.
6. A formula sheet is attached at the end of this question paper.
7. Round off answers to TWO decimal places, unless stated otherwise.
8. Write neatly and legibly.

QUESTION 1: DC MACHINES

1.1 A 4-pole, lap-wound DC generator has 720 armature conductors and a flux of 25 mWb per pole. Calculate the generated EMF when it runs at 1 000 r/min. (4)

1.2 A 220 V DC shunt motor has an armature resistance of 0,4 Ω and a field resistance of 110 Ω. At full load it takes a line current of 32 A and runs at 1 000 r/min.

1.2.1 Calculate the field current and the armature current. (2)

1.2.2 Calculate the back EMF. (2)

1.2.3 Calculate the gross torque developed by the armature. (3)

1.2.4 The load is increased so that the armature current becomes 50 A. Assuming that the flux remains constant, calculate the new speed. (3)

1.3 For the motor in QUESTION 1.2 at full load, the rotational (iron and friction) losses are 500 W. Calculate:

1.3.1 the armature copper loss and the field copper loss (2)

1.3.2 the output power and the efficiency (4)

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QUESTION 2: THREE-PHASE CIRCUITS AND POWER FACTOR CORRECTION

2.1 A balanced star-connected load with an impedance of (12 + *j*16) Ω per phase is connected to a 400 V, 50 Hz, three-phase supply. Calculate:

2.1.1 the phase voltage and the line current (4)

2.1.2 the power factor (2)

2.1.3 the active, reactive and apparent power (4)

2.2 A factory takes 50 kW at a power factor of 0,7 lagging from a 400 V, 50 Hz, three-phase supply. The power factor must be improved to 0,95 lagging with three delta-connected capacitors.

2.2.1 Calculate the reactive power that the capacitors must supply. (5)

2.2.2 Calculate the capacitance of each capacitor. (3)

2.2.3 Give TWO advantages of power factor correction for the factory. (2)

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QUESTION 3: TRANSFORMERS

3.1 A 100 kVA, 11 000/400 V, 50 Hz, single-phase transformer has 80 turns on the secondary winding. Calculate:

3.1.1 the number of primary turns (2)

3.1.2 the maximum value of the flux in the core (3)

3.2 The open-circuit test on the transformer gave a power reading of 1,2 kW, and the short-circuit test at full-load current gave 1,8 kW. Calculate the efficiency:

3.2.1 at full load, power factor 0,8 lagging (4)

3.2.2 at half load, unity power factor (4)

3.2.3 Calculate the kVA load at which maximum efficiency occurs. (4)

3.3 Explain why transformers are rated in kVA and not in kW. (3)

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QUESTION 4: THREE-PHASE INDUCTION MOTORS

4.1 A 4-pole, 50 Hz, three-phase induction motor runs at a slip of 4% at full load. Calculate:

4.1.1 the synchronous speed (2)

4.1.2 the rotor speed (2)

4.1.3 the frequency of the rotor currents (2)

4.2 The motor takes 15 kW from the supply. The stator losses are 0,8 kW and the friction and windage losses are 0,3 kW. Calculate:

4.2.1 the rotor input (air-gap) power (2)

4.2.2 the rotor copper loss (2)

4.2.3 the output power and the efficiency (3)

4.2.4 the shaft torque (3)

4.3 Explain how a star-delta starter reduces the starting current of an induction motor, and state by what factor the line current is reduced. (4)

[20]

QUESTION 5: SYNCHRONOUS MACHINES

5.1 A 2 MVA, 11 kV, three-phase, star-connected alternator has a synchronous reactance of 10 Ω per phase. The armature resistance can be ignored. The alternator supplies full load at a power factor of 0,8 lagging.

5.1.1 Calculate the phase voltage and the full-load current. (3)

5.1.2 Calculate the EMF per phase that must be generated. (5)

5.1.3 Calculate the percentage voltage regulation. (2)

5.1.4 Draw the phasor diagram for this condition. (2)

5.2 Name FOUR conditions that must be met before an alternator is connected in parallel with the grid. (4)

5.3 Explain the effect of increasing the field excitation of a synchronous motor on its power factor, and state ONE practical application of this effect. (4)

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TOTAL: 100

END OF QUESTION PAPER

FORMULA SHEET

$E = \frac{Z\Phi Np}{60a}$ (lap: a = p; wave: a = 2)

$E_b = V - I_aR_a$

$T = \frac{E_bI_a}{2\pi N/60}$

$\frac{N_2}{N_1} = \frac{E_{b2}\Phi_1}{E_{b1}\Phi_2}$

$\eta = \frac{\text{output}}{\text{input}} \times 100\%$

Star: $V_L = \sqrt{3}\, V_{ph}$; $I_L = I_{ph}$

Delta: $V_L = V_{ph}$; $I_L = \sqrt{3}\, I_{ph}$

$P = \sqrt{3}\, V_L I_L \cos\varphi$; $Q = \sqrt{3}\, V_L I_L \sin\varphi$; $S = \sqrt{3}\, V_L I_L$

$Q = P \tan\varphi$; $X_C = \frac{1}{2\pi fC}$; $Q_C = \frac{V^2}{X_C}$

$E = 4{,}44\, f N \Phi_m$; $\frac{V_1}{V_2} = \frac{N_1}{N_2}$

$\eta = \frac{x S \cos\varphi}{x S \cos\varphi + P_i + x^2 P_{cu}}$; $x_{\max\eta} = \sqrt{\left(\frac{P_i}{P_{cu}}\right)}$

$N_s = \frac{120f}{p}$; $s = \frac{N_s - N_r}{N_s}$; $f_r = sf$

Rotor copper loss = $s \times$ rotor input; $P_{mech} = (1 - s) \times$ rotor input

$T = \frac{P}{2\pi N/60}$

$E = V + I(R + jX_s)$; regulation $= \frac{E - V}{V} \times 100\%$